DEXATHON 2026 · FINAL · TEAM STERNRITTERS (DEX26302)

NEERKATTI

An AI water-keeper for Chennai's wetlands



28.1 ha
wetland flagged as built over

36
patches, 2018 to 2026

₹0
to run, no server

athulyapandian24.github.io/neerkatti/app/?final=1

Two tiers, split on purpose

1

Offline pipeline

Google Earth Engine + Python.
Nine dry seasons of Sentinel-2,
labelled square by square.

2

Static artefacts

GeoJSON, one data file,
WebP image layers.
Built once, not per request.

3

Zero-backend client

Leaflet + plain JavaScript
on GitHub Pages.
A file fetch, nothing more.

All cost moves to build time. The map and the evidence need no server, no database and no API key — nothing to hack, nothing to bill, nothing to go down during judging.

The one exception, live: the citizen photo check calls Gemini 2.5 Flash-Lite from a serverless function, with the key held as a secret the browser never sees. It is an add-on — the map works without it.

Every component earns its place

Sentinel-2
Free, 10-metre, 5-day revisit — the only imagery that makes a nine-year record affordable

Dynamic World
A validated AI land-cover model, so we don't train a weak one on no ground truth

Earth Engine
Moves compute to the data — petabytes processed without downloading a single scene

Leaflet 1.9.4
38 KB, vendored into the repo rather than a CDN: no third-party dependency at runtime

No framework
Plain JavaScript keeps the payload small and the app alive on a cheap phone

Where the engineering actually is

1

Client-side AI lookup

The AI label layers are lossless WebP — one exact colour per class. Tap anywhere and the browser reads that single pixel off a canvas and reports what the AI saw there in every season.

No inference server. No round-trip.

2

A free comparison

The 2018 | 2026 divider is a CSS clip on a Leaflet pane, so both years share one map and one set of tiles — no second instance to keep in step.

Nothing extra to download.

3

One published API

Every final-round feature is its own file talking to window.NEERKATTI_API — alerts, the letter, the whole new interface.

Four people, no conflicts in the map core.

Built for a cheap phone on slow 4G

600 px

first image, full one only on zoom

100%

works offline after one visit

₹0

hosting, compute and data

- The light copy of each season loads first; the full-resolution image is fetched only when the screen can actually show more detail.
- The year player preloads the next seasons, so 2018 to 2026 plays without a stall.
- A service worker precaches every season's imagery; the map tile cache is capped at 400 so storage never runs away.
- Everything is pre-computed, so there is no query, no cold start and no rate limit at demo time.

62 automated tests, on every push

62

43 consistency tests
19 map-logic tests

Green in CI on every commit

The tests assert

- every number printed on the page matches the analysis output
- no script loads from a CDN, and the offline list points at real files
- the label images stay lossless and identically sized — the pixel lookup depends on it

And it degrades gracefully

- map library blocked → the patch list and the data still work
- a full image fails → its light copy stays on screen
- no connection → an offline banner and the saved copy of the map

Rejecting what a naive diff would call a change

26

raw detections

Seasonal

bare or built in the same months last year — a lake edge that dries every summer

Already built

built in the 2025 dry season: an old change, not a new one

Moving shoreline

within 30 m of water the AI still sees — filling takes the water away

5

possible

Emerging technology, used honestly

AI land-cover labels read in the browser, with no model server.

Automated complaint letters in Tamil and English, generated from the patch data and the dated satellite close-up.

Technical risk we refuse to hide

Every patch and alert says “possible change, needs a ground check”.

We never name an owner and never call anything illegal. The top 10 were checked by eye: 9 real, 1 unclear — and we publish the unclear one.

Free data. Free compute. Free hosting.

Fully tested, and it scales to any water body in Tamil Nadu without new cost.

28.1 ha	62	₹0
flagged, 65% of it since 2023	tests green on every push	to run, today and at scale

athulyapandian24.github.io/neerkatti/app/?final=1

NEERKATTI · Team Sternritters (DEX26302) · Thank you